Skills Summary

Summarizing Distributions with Center, Spread and Shape

Use this reference while completing your worksheet or when studying.

Skill 1 — Measures of center

Mean = $\dfrac{\text{sum of the values}}{\text{how many values}}$ — the balance point, which every value pulls on. **Median** = the middle value of the sorted list (the average of the two middle values when the count is even) — a position, not a total.

Example: Find the mean of $3, 5, 8, 10, 14$.
Step 1. The mean shares the total equally, so add everything and divide by how many values there are.
Step 2. $\dfrac{3+5+8+10+14}{5} = \dfrac{40}{5}$.
$\Rightarrow$ mean = **8**

Try it: Find the mean of $2, 4, 9, 11, 14$.

check: **8**

Skill 2 — Measures of spread

Range = largest − smallest: one number, easily wrecked by a single extreme value. **IQR** $= Q_3 - Q_1$, the width of the middle half. Split the sorted list at the median, then take the median of each half. **Standard deviation** is the typical distance of a value from the mean.

Example: Find the IQR of $4, 6, 8, 11, 15, 18, 20, 22$.
Step 1. The IQR describes the middle half, so split the sorted list in two and find the median of each half.
Step 2. Lower half $4, 6, 8, 11$ has median 7; upper half $15, 18, 20, 22$ has median 19; the difference is $19 - 7$.
$\Rightarrow$ IQR = **12**

Try it: Find the IQR of $1, 3, 6, 8, 12, 14, 17, 19$.

check: **11**

Skill 3 — Shape, outliers, and which center to trust

Rule: compare the mean with the median. Mean above median means a right tail (right-skewed); mean below means a left tail; roughly equal means roughly symmetric. An outlier is a value beyond $Q_1 - 1.5\,\text{IQR}$ or $Q_3 + 1.5\,\text{IQR}$. The median and the IQR are *resistant* — an extreme value barely moves them — so report those when a distribution is skewed or has outliers.

Example: Find the median of $5, 6, 7, 8, 40$.
Step 1. The 40 will drag the mean upward, so the median is the honest center here — take the middle value of the sorted list.
Step 2. Five values, so the middle one is the third: $5, 6, \underline{7}, 8, 40$. (The mean, by contrast, is 13.2.)
$\Rightarrow$ median = **7**

Try it: Find the median of $2, 3, 4, 5, 36$.

check: **4**

Skill 4 — Reading a display

Rule: a frequency table, bar chart or dot plot stores counts, not the original list. The total is the sum of the counts; the tallest bar names the most common category. Translate the display into the notation before computing: "how many altogether" is a sum, "how many more" is a difference, "what fraction" is a count over the total.

Example: A tally of favourite colours reads Red 8, Blue 14, Green 6. How many were surveyed?
Step 1. Each person is counted in exactly one category, so the number surveyed is the sum of the counts.
Step 2. $8 + 14 + 6$.
$\Rightarrow$ total surveyed = **28**

Try it: A pet tally reads Cats 9, Dogs 17, Fish 4. How many pets altogether?

check: **30**